Lecture 4: Simple regression

Empirical Methods for Law and Economics | EDLE

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Today's question

Italy takes 1,120 days to enforce a contract.

Is that slow *for an economy as rich as Italy*?

- By the end of today you will be able to:
  - merge two datasets and check what was lost;
  - fit a simple regression and interpret the slope in units, including with logarithms;
  - use predictions and residuals as a benchmark, and say what a residual is not;
  - read a regression on a $0/1$ variable as a comparison of means;
  - read one coefficient of a published table.

DISCUSS Slow compared to what?

- Italy: 1,120 days. Median of 191 economies: 575. Greece: 1,711. France: 447. Germany: 499.

Propose a benchmark for "slow", and say what it assumes.

- The world median? The EU average? Italy's own past? Economies with similar income?

ANSWERS Slow compared to what?

- Every benchmark is a comparison group, and every comparison group carries an assumption:

- world median: Italy should look like all 191 economies; EU average: membership is what matters; Italy's past: only time changes; economies with similar income: income sets what courts can do.

- A benchmark that depends on income is a **prediction from a line**: a regression.

STATA Guided Stata 1: bring in income

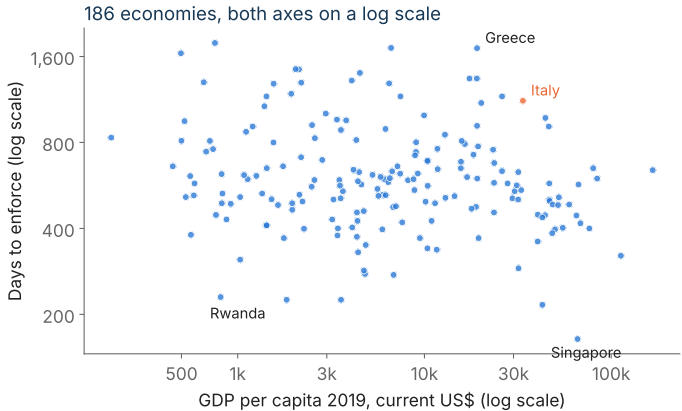
```
use "$course_data/contracts_2019.dta", clear
merge 1:1 economy_code using "$course_data/gdp_2019.dta"
tabulate _merge
list economy if _merge == 1
```

Matched (in both files)	186
Only in Doing Business (_merge==1)	5
Only in the income file (_merge==2)	24

- No GDP for 2019 (as listed by Stata): Eritrea; South Sudan; Taiwan, China; Venezuela, RB; Yemen, Rep.

Income file: World Bank, World Development Indicators, GDP (current US$) and population, 2019, CC BY 4.0; GDP per capita computed as their ratio.

PREDICT Income and enforcement time



Predict: the sign of the line, and how much of the variation it will explain.

The regression line

$$y_i = \beta_0 + \beta_1 x_i + u_i \qquad \text{OLS: choose } \hat{\beta}_0, \hat{\beta}_1 \text{ to minimise } \sum_i \hat{u}_i^2$$

- $\hat{\beta}_1 = r_{xy} \dfrac{s_y}{s_x}$: the correlation, rescaled into the units of y per unit of x.
- Prediction: $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$. Residual: $\hat{u}_i = y_i - \hat{y}_i$.
- The line is the best linear approximation to the **conditional mean** of y given x. Whether β_1 is a causal effect depends on the design, not on the method.

Reading slopes when logarithms are involved

Model	Slope $\hat{\beta}_1$ means	Example
y on x	$+1$ unit of $x \Rightarrow \hat{\beta}_1$ units of y	days per point of the quality index
y on $\ln x$	$+1\%$ in $x \Rightarrow$ about $\hat{\beta}_1/100$ units of y	days per 1% more income
$\ln y$ on x	$+1$ unit of $x \Rightarrow$ about $100\hat{\beta}_1\%$ in y	percent change in days per index point
$\ln y$ on $\ln x$	$+1\%$ in $x \Rightarrow$ about $\hat{\beta}_1\%$ in y	elasticity of time with respect to income

- The approximations hold for small changes. For large ones compute exactly: doubling x in a log-log model multiplies y by $2^{\hat{\beta}_1}$.

STATA Guided Stata 2: fit the line

```
keep if _merge == 3
generate ln_time  = ln(time_days)
generate ln_gdppc = ln(gdp_pc_usd)
regress ln_time ln_gdppc
```

	Coefficient	Std. error	t	p-value
ln_gdppc	-0.049	0.022	-2.27	0.025
constant	6.819	0.193	35.41	< 0.001
$n = 186$,	$R^2 = 0.027$			

- 10% higher GDP per capita: about 0.5% fewer days. Doubling income: $2^{-0.049} = 0.966$, about 3.4% fewer days.

DISCUSS "$R^2 = 0.03$: income explains nothing about courts"

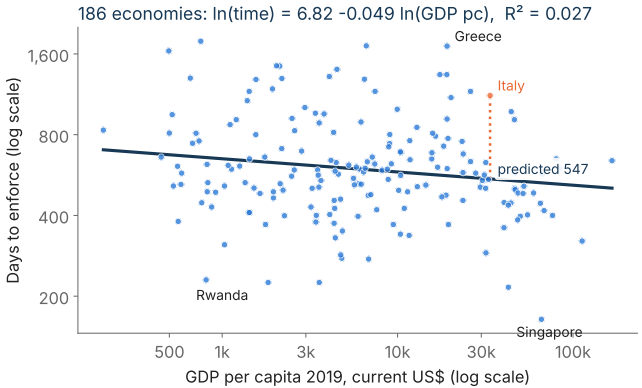
Agree or disagree, with one argument each.

1. The slope is statistically different from zero ($p = 0.025$). How can both statements be true?
2. Is a 3% reduction in time for a doubling of income large or small for a policy maker?
3. What else could explain the other 97%?

ANSWERS "$R^2 = 0.03$: income explains nothing"

- Both can be true: R^2 measures how much of the variation the line accounts for; the p-value tests whether the slope differs from zero.
- Doubling income goes with about 3.4% fewer days: roughly 20 days on a 600-day case. Small.
- The other 97%: procedure, legal tradition, judges and their organisation, litigation volume, measurement error in a standardized case.
- R^2 is about fit, not about importance or causality.

Prediction and residual for Italy



- Italy: predicted $e^{6.305} = 547$ days; observed 1,120. Residual $= \ln 1120 - 6.305 = 0.716$: Italy takes $e^{0.716} = 2.05$ times the *typical* time predicted by its income.

DISCUSS Is Italy's residual a measure of inefficiency?

Questions:

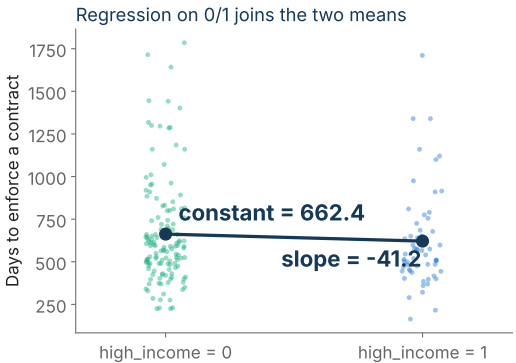
1. List what is inside the residual: everything that affects time and is not GDP per capita.
2. Greece has the largest positive residual, Singapore the largest negative one. Would you rank court systems by residual?
3. What would change if the benchmark were judicial quality instead of income?

ANSWERS Is Italy's residual a measure of inefficiency?

- Inside the residual: procedure, organisation, judges' workload, litigation culture, the specific standardized case, measurement error.
- Ranking by residual ranks countries by everything the model leaves out, measured with error.
- A different benchmark (e.g. judicial quality) gives a different prediction and a different residual.
- A residual is the start of a question, not an answer.

STATA Guided Stata 3: a 0/1 regressor

```
use "$course_data/contracts_2019.dta", clear      // all 191 economies
generate byte high_income = (income_group == "High income")
regress time_days high_income
ttest time_days, by(high_income)
```



- Constant = 662.4: mean of the 0 group.
- Slope = −41.2 (SE 47.7, $p = 0.389$): mean of group 1 minus mean of group 0.
- Same t and p as yesterday's `ttest`, with the opposite sign of the difference.

Uncertainty about a slope

- The coefficient is an estimate: another set of economies or another year would give another value.
- Its standard error plays the same role as the standard error of a mean in Lecture 2.
- $t = \hat{\beta}_1 / \mathrm{SE}(\hat{\beta}_1)$ tests H_0: $\beta_1 = 0$. The 95% interval is roughly $\hat{\beta}_1 \pm 2\,\mathrm{SE}$.
- Income slope: -0.049, interval -0.092 to -0.006. Quality index slope (all 191): -20.7 days per point, interval -33.8 to -7.7.
- These standard errors rest on assumptions (independent observations, constant variance) that we examine in Lecture 6.

DISCUSS Paper: Voigt and Wulf (2019), data and measures

- Question: which institutions make prosecutors independent from government *in practice*?
- One observation = one **country**; 76 countries in the data, 74 in column (1) of Table 2. One cross-section.
- Outcome: *de facto* independence, an index from 0 to 1 built from up to 7 items on actual practice (e.g. prosecutors transferred or retired against their will). Mean 0.52.
- *De jure* independence: index from 0 to 1, up to 21 items read in legal texts (e.g. can the executive give instructions?). Mean 0.46.
- The two are weakly *negatively* correlated ($r = -0.21$, Van Aaken et al. 2010).

Why might the law on paper and the practice diverge? Which one would you trust as a measure, and why?

Voigt, S. and Wulf, A.J. (2019), *Journal of Institutional Economics* 15(1): 99–120, Section 5 and Table 1.

- *De jure* independence may not bind: laws can be ignored, and informal pressure needs no legal power.

- *De facto* independence is closer to what matters, but depends on what is observed and reported: a forced transfer must be known to be counted.

- The weak negative correlation between the two ($r = -0.21$) is the reason the paper studies *de facto* independence.

DISCUSS Reading column (1) of Table 2

Dependent variable: de facto index	(1)
De jure independence	−0.20 (0.31)
Press freedom	0.01* (0.01)
Polity	−0.01 (0.01)
Legal origin: English	0.24* (0.09)
Legal origin: French	0.16† (0.09)
GDP p.c. (thousand US$)	0.01 (0.01)
Constant	0.73** (0.24)
R^2 / N	0.29 / 74

Standard errors in parentheses. **1%, *5%, †10%. Reference legal origin: socialist, German, Scandinavian.

Questions:

1. **English 0.24**: a 0/1 regressor, as in Guided Stata 3. Compared with which countries? Write the sentence.
2. **Press freedom 0.01 (0.01)**, yet significant at 5%. What does two-decimal rounding hide? The authors write that a 10-point increase "*leads to*" a 0.10 rise: is "leads to" justified?
3. Column (1) has six regressors: every coefficient holds the other five fixed. Preview of Lecture 5.
4. Footnote 10: a non-significant normality test "indicat[es] that our data are from a normally distributed population". Compare with "$p = 0.39$: income does not matter".

ANSWERS Reading column (1) of Table 2

- **English 0.24** (SE 0.09): English legal-origin countries score 0.24 higher on the 0–1 index than socialist, German or Scandinavian legal-origin countries, holding the other five variables fixed. Legal origin is not randomly assigned.

- **Press freedom 0.01 (0.01)**: the printed digits are too rounded to recover t. "Leads to" is causal language for a cross-country association; press freedom correlates 0.71 with GDP per capita (fn. 11).

- **Six regressors**: each coefficient holds the other five fixed (Lecture 5).

- **Footnote 10**: not rejecting normality does not show normality, the same error as reading $p = 0.39$ as "no difference".

Laboratory

- Worksheet: `do/exercises/04_simple_regression.do`. Lab Companion, Lab 4.
- Data: `contracts_2019.dta` merged with `gdp_2019.dta` (Tasks 1–3); `contracts_2019.dta` alone (Tasks 4–5); `justice_europe_2019.dta` (Task 6).
- Every slope written as a sentence with units; every residual described as a question, not a verdict.

Exit check

1. Write the income slope (-0.049, log-log) as a sentence with units.
2. Italy's residual is 0.716. Give one thing it might measure other than inefficiency.
3. A regression of days on a $0/1$ high-income variable has slope -41.2. What two numbers does it compare?

Next week

- Lecture 5: multiple regression. Back to police and theft: what happens to the police coefficient when we hold income constant?
- Before meeting 5: one-page outline of your referee report (question, data, method).